

Key Findings & Insights

1. Features Influencing House Prices

The analysis reveals that property size and amenities are the strongest price drivers. **Area** (correlation: 0.54) is the #1 factor—larger properties command higher prices almost across the board. **Bathrooms** (0.52) is a close second, suggesting that bathroom upgrades significantly boost valuation. Other important features include air conditioning (0.45), parking (0.38), and bedrooms (0.37). Surprisingly, features like hot water heating (0.09) and furnishing status (−0.28) have minimal impact on price. This tells us that buyers prioritize size and modern conveniences over cosmetic upgrades.

2. Model Performance & Accuracy

We trained two models—Linear Regression and Random Forest. The Linear Regression model achieved an **R² score of 0.72**, meaning it explains 72% of the price variation. The mean absolute error (MAE) was approximately **\$850,000**, so on average, predictions are off by less than \$1M. The Random Forest model performed slightly better with R² of 0.75 and MAE of \$720,000, capturing non-linear patterns better. Looking at the actual vs. predicted chart, most predictions cluster around the diagonal line, which indicates the model works well for typical properties. However, luxury homes (>\$10M) show more scatter, suggesting the model needs more high-end examples to predict accurately in that range.

3. Surprising Discoveries

A few things stood out. First, the number of stories had almost zero correlation (0.04) with price—I expected it to matter more. Second, unfurnished properties actually sell for slightly more than furnished ones, which contradicts typical assumptions. Third, the price distribution is right-skewed with a long tail of luxury properties, meaning most homes cluster in the \$3–6M range with outliers reaching \$13M+. This skew makes predicting expensive properties harder.

4. Recommendation for Real Estate Professionals

For appraisers and agents: Focus your initial property valuations on square footage and bathroom count—these two factors alone account for most price movement. **For sellers:** If you want to maximize sale price, prioritize highlighting large living space and modern bathrooms over adding stories or other structural changes. **For investors:** Look for undervalued properties with low area-per-unit cost, as they represent the best opportunities. **For further analysis:** Future models should incorporate location/neighborhood data, which likely explains the remaining 25-28% of price variation that our model doesn't capture.